

Sensiq

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Abstract—
Index Terms—

I. INTRODUCTION

Reliable indoor environment quality (IEQ) monitoring is a recurring requirement in laboratories, production halls, and similar facilities, where ambient conditions like indoor air quality (IAQ) directly affect experimental validity, product quality, and occupational safety. Existing Internet of Things (IoT) monitoring platforms address parts of this problem but typically rely on on-premises servers, offer only coarse sampling, or restrict access through proprietary applications (Section II).

This report describes Sensiq, a microcontroller unit (MCU)-based IoT architecture combining ESP32 (Espressif Systems) edge devices with a serverless Amazon Web Services (AWS) cloud backend. Sensor readings are transmitted securely over Message Queuing Telemetry Transport (MQTT) using Transport Layer Security (TLS), validated, and persisted for both live and historical views. Exposed through Representational State Transfer (REST) Application Programming Interface (API) endpoints, a React-based frontend visualises the data, while threshold-driven email alerts notify personnel of critical events.

A. Mission Statement

Sensiq aims to deliver an end-to-end, low-latency environmental monitoring system that provides both real-time visibility and cost-efficient historical analytics. By leveraging a fully serverless backend, it enables stakeholders to continuously track critical ambient parameters—such as temperature, humidity, light intensity, gas concentration, and flame exposure—to ensure safety and quality assurance.

B. Motivation

Unnoticed deviations in ambient conditions can invalidate experiments, damage sensitive samples, and endanger personnel [1, 2]. While continuous monitoring is essential, existing academic IoT platforms (e.g., [3, 4]) often rely on high-maintenance on-premises stacks and employ sampling intervals that are too coarse to capture short-lived events. Sensiq closes this gap by pairing inexpensive edge hardware with a scalable cloud architecture, keeping infrastructure and operating costs low enough for routine laboratory use. The full source code is publicly available under the MIT License [5].

C. Document Structure

The remainder of this report is organised as follows. Section II positions Sensiq against existing academic and commercial monitoring solutions and motivates the design decisions taken in this project. Section III introduces the conceptual building blocks of the system and presents the end-to-end architecture in a technology-agnostic manner.

The subsequent three sections refine this concept along the data flow: Section IV details the ESP32-based sensor node and its secure MQTT publication to AWS IoT Core; Section V describes the serverless cloud pipeline, covering both the low-latency live route and the long-term history route; and Section VI explains how end users consume the data through the React-based web frontend and how they are notified of critical events via email. Section VII assesses the resulting system with respect to testability and operating cost. Finally, Section VIII summarises the contribution of the report and outlines directions for future work.

II. RELATED WORK

Several IoT-based platforms for indoor environmental monitoring have been proposed. This section contrasts two representative academic systems — by Mumtaz et al. [3] and Marques and Pitarma [4] — with the design decisions taken for Sensiq.

A. On-Premises IoT Air Quality Monitoring

Mumtaz et al. [3] present a multi-sensor node targeting air quality prediction. While sharing Sensiq’s general topology, their system differs fundamentally in three areas.

First, they rely on a local web server and database, tying the platform to on-premises operation. Sensiq instead targets a fully serverless cloud backend (Section V) to scale on demand and keep idle costs near zero.

Second, their five-minute sampling interval is geared towards trend analysis, which is too coarse for safety-relevant events like fires or for product quality assurance where short-lived deviations can compromise samples. Sensiq utilizes a low-latency MQTT-over-TLS pipeline to surface critical conditions immediately.

Third, their twenty-hour battery autonomy limits remote deployment. Sensiq’s ESP32 architecture supports deep sleep operation, planned as a future extension (Section VIII).

B. iAQ+ Laboratory Environment Supervision System

Marques and Pitarma [4] propose iAQ+, an ESP8266-based monitoring system. While hardware choices are similar, significant differences exist in backend architecture, frontend access, and timing guarantees.

On the backend, iAQ+ employs a traditional server-based stack (SQL Server and .NET), which requires continuous hosting. Sensiq delegates persistence and routing to managed AWS services.

On the frontend, iAQ+ restricts access to iOS devices and limits the web dashboard to hourly aggregations. Sensiq provides platform-independent access via open REST endpoints, a React frontend, and a true real-time view (Section VI).

Finally, iAQ+ lacks explicit latency guarantees for threshold alerts. Sensiq is efficiently designed around instantaneous, threshold-driven email notifications.

III. TECHNICAL CONCEPT

High-level, technology-agnostic view of the system. This section introduces the conceptual building blocks and the end-to-end data flow. Detailed component-level descriptions follow in the Hardware, AWS Services, and User Interaction sections.

A. Technical Building Blocks

Enumerate the logical components of the system without binding them yet to concrete products: edge sensing node, secure transport channel, ingestion gateway, validation stage, hot-path store (live data), cold-path store (history), query API, notification channel, and user interface. Briefly justify the chosen programming languages (C++ firmware, Python lambdas, TypeScript infrastructure, React UI).

B. System Architecture

Present the architecture diagram (see template: fig:cloud-architecture) and walk the reader through the data flow: ESP32 -> MQTT/TLS -> AWS IoT Core -> validation Lambda -> dual persistence (DynamoDB for live, S3 + Athena for history) -> REST endpoints -> frontend / SNS.

IV. HARDWARE

The sensor node is the physical edge component of the Sensiq system, built around an ESP32 microcontroller with a set of environmental sensors. This section maps the hardware components (Section IV-A), outlines the firmware measurement loops (Section IV-B), and details the secure MQTT publication strategy (Section IV-C).

A. Component Overview

The sensor node is built using the following core components:

- **ESP32 NodeMCU:** Central microcontroller with integrated Wi-Fi and Bluetooth [6, 7].
- **DHT11:** Basic digital temperature and humidity sensor [8].
- **Joy-it KY-026 & KY-028:** Analog flame sensor (IR band) and NTC thermistor, both with adjustable digital thresholds [9, 10].

- **TSL2561:** Digital ambient-light sensor for visible and IR light [11].
- **Bosch BME680:** Comprehensive sensor for precise temperature, humidity, pressure, and VOC gas concentrations [12].
- **Auxiliary Switches:** Two inputs reserved for labeling training data and outliers (see Section VIII).

The hardware design is documented using two complementary views. Figure 1 shows the logical circuit diagram (created with KiCAD [13]), detailing wiring and GPIO assignments. The functional prototype is shown in Figure 2, demonstrating the assembled sensor node.

B. Measurement Workflow

The firmware on the ESP32 follows a deterministic, two-phase lifecycle written in Arduino C++ [14]: a one-time *startup* phase that prepares the networking and sensing subsystems, followed by a continuously running *measurement loop*.

During startup the node performs the following steps in order:

- 1) Wi-Fi association: The node connects to a configured local network using a pre-provisioned SSID and password.
- 2) Time synchronization: The internal clock is synchronized with an NTP server. An accurate wall-clock time is required both for meaningful measurement timestamps and for the validation of TLS certificates during the subsequent MQTT handshake.
- 3) Sensor initialization: GPIO pin modes are set and sensor-specific configuration (e.g. DHT11 configuration) is applied.
- 4) MQTT setup: The MQTT client is configured (see Section IV-C).

Once startup is complete, the node enters its measurement loop:

- The MQTT connection is continuously validated and kept alive; on disconnect, the client transparently reconnects.
- Every t milliseconds a full sensor reading is taken across all attached sensors with a default waiting period of $t = 1000$ ms. Values smaller than one second might not change much, as the sensors often have a minimum sampling rate of 1 Hz or lower.
- After n consecutive readings the buffered samples are aggregated into a single record, serialized as JSON and published to the configured MQTT topic. The standard sample aggregation size is $n = 5$, guaranteeing a balance between responsiveness and noise reduction.

Aggregation reduces sensor noise and the influence of isolated outliers. It is performed per field according to its data type: the most recent timestamp of the buffer is taken as the record timestamp, numerical values (analog readings) are averaged arithmetically, and digital boolean signals are decided by majority voting over the buffered samples.

For the KY-028 module, the raw 12-bit ADC reading is first converted to a resistance and then mapped to an absolute temperature in degrees Celsius using the Steinhart-Hart equation [0].

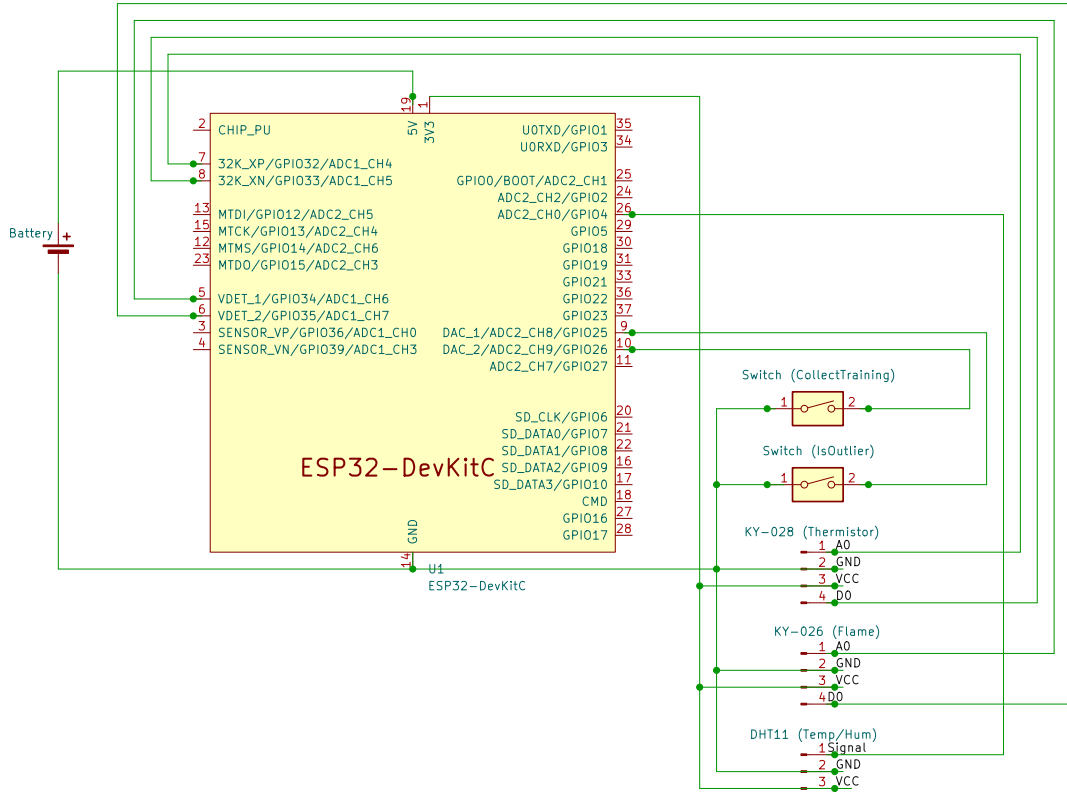


Figure 1. Circuit diagram of the sensor node, showing the logical wiring between the ESP32-DevKitC, sensors, the two auxiliary switches and the power supply, together with their respective GPIO assignments.

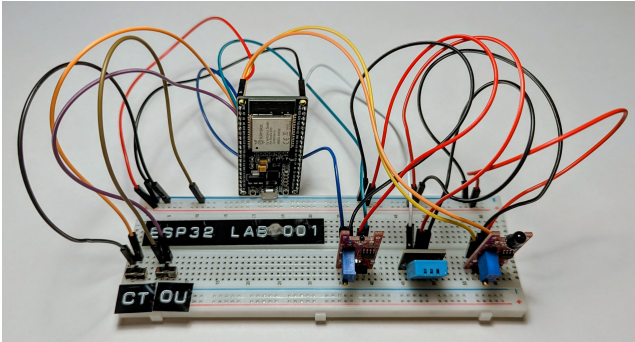


Figure 2. Real physical sensor node with the assembled sensors, the ESP32-DevKitC and the auxiliary switches.

C. MQTT Publication to AWS IoT Core

The sensor node communicates with AWS IoT Core via the MQTT protocol over TLS.

To establish a secure connection to the AWS IoT broker on port 8883, the node requires the broker address and a set of cryptographic credentials. The publicly available AWS Root CA certificate alongside the individual device certificate and device private key are used for mutual TLS authentication.

The MQTT topic structure is designed around a unique device identifier (e.g., esp32-lab-001). Data exchange uses

```

1 {
2     "running_time": 9881892,
3     "timestamp": "2026-05-25T22:56:12Z",
4     "device_id": "esp32-lab-001",
5     "location": "Lab A, OTH Amberg-Weiden, 92224 Amberg,
6         Germany",
7     "dht_humidity": 61,
8     "dht_temperature": 23.8,
9     "dht_heat_index": 23.82809,
10    "flame_analog": 0,
11    "flame_digital": false,
12    "thermistor_analog": 2005,
13    "thermistor_digital": false,
14    "thermistor_temp": 24.05634,
15    "is_outlier": false,
16    "collect_training": false
17 }

```

Listing 1. Published JSON payload by the ESP32 sensor node via MQTT.

three primary topics: sensiq/<identifier>/data for publishing the aggregated sensor records, sensiq/<identifier>/test for verifying connectivity, and sensiq/<identifier>/commands reserved for receiving subsequent control commands. Listing 1 shows an

example of the JSON payload that the ESP32 node publishes to the data topic after each aggregation interval.

V. AWS SERVICES

Detailed description of the cloud-side processing pipeline. Two functional routes are exposed by the backend: a low-latency live route and a long-term history route. Each is described in its own subsection together with the AWS services it relies on.

A. Live Route

Path from IoT Core rule -> validation Lambda -> DynamoDB (latest reading per device) -> API Gateway -> GET /live. Discuss latency targets, payload shape, and error handling.

B. History Route

The history route provides a long-term data retrieval mechanism, exposing a GET /history endpoint via API Gateway. It accepts query parameters such as time ranges and limits, and returns historical sensor data as a structured JSON payload. The pipeline bridges data ingestion from AWS IoT Core to a highly optimized query backend.

1) *Data Ingestion and Storage:* Incoming JSON messages from AWS IoT Core are captured by an IoT rule and forwarded to Amazon Kinesis Data Firehose [0]. To mitigate the small-files problem, Firehose buffers the data for up to 15 minutes or until a size of 128 MB is reached before converting it to Apache Parquet format. The Parquet files are written to a S3 Standard data bucket [0], partitioned by year, month, and day to ensure efficient storage and rapid retrieval. S3 lifecycle management policies [0] are configured to retain the data securely for an extended period, while an AWS Glue Data Catalog maintains the structural metadata of the `sensor_data` table. IAM policies tightly control access to the Glue catalog and underlying S3 buckets [0].

2) *Query Execution and Processing:* A dedicated AWS Lambda function coordinates the fulfillment of history requests. It parses the incoming query parameters, validates them, and constructs an Athena query using query execution parameters [0] to defend against SQL attacks. Athena utilizes Glue's partition projection mapping [0] to prune directories rapidly without needing to load external metadata. A dedicated Athena workgroup [0] manages the query execution, saving the result datasets to a secondary S3 query results bucket.

During execution, the Lambda function continually polls the query execution status [0]. Once the query yields a terminal state, the Lambda handler retrieves the generated results from the S3 bucket, formats them into a JSON response, and returns them to the API Gateway. Although Athena queries induce higher latency compared to the live route, processing is typically completed well within the 29-second timeout limit, mostly in under 5 seconds.

VI. USER INTERACTION

Describes how end users consume the data and receive alerts.

A. Email Notifications via SNS

Explain the alert pipeline: threshold evaluation in the validation Lambda, publication to an SNS topic, and email delivery to subscribed recipients. Cover debounce behaviour and message content.

B. Web Frontend

Describe the React-based dashboard hosted on AWS Amplify: live tiles, historical charts, sensor status indicators, and authentication via AWS Cognito. Include a screenshot or mock-up (see template: figure*).

VII. EVALUATION

A. Testability

The cloud system's testability is ensured through rigorous automated checks for both infrastructure and application logic. Local yarn scripts automate test execution and coverage reporting.

1) *AWS Infrastructure:* The AWS cloud infrastructure configuration is automatically validated using built-in AWS Cloud Development Kit (CDK) assertions, leveraging the `Template` and `Match` constructs. This ensures that the generated resources and policies match the intended architectural design prior to deployment.

2) *AWS Lambda Handlers:* AWS Lambda functions are thoroughly unit-tested using Python's `unittest` or `pytest` frameworks, employing exhaustive mocking to isolate execution behavior. To evaluate Lambda handlers independently of the API Gateway, a locally defined `TestEvent` JSON payload is utilized to replicate realistic incoming requests during testing. Additionally, for runtime observance and debugging in the deployed environment, Lambda function executions and errors are continually logged to dedicated AWS CloudWatch log groups.

3) *Sensor Node Firmware:* Firmware unit testing utilizes the AUnit framework [18] to cover modules like sensor readings, data aggregation, configurations, and Wi-Fi. Exhaustive MQTT mocking was omitted due to real-world deviations. Instead, the node subscribes to an MQTT test topic during operation to directly verify connection health and publish functionality.

B. Cost Estimation

1) *AWS Service Costs:*

2) *Hardware Cost:* The one-time hardware cost per sensor node is based on the components used for prototyping. A breakdown of the individual component costs is presented in Table I.

VIII. SUMMARY AND FUTURE WORK

Recap the contribution of the report: a working, MCU-based IoT architecture for environmental monitoring with secure ingestion, dual-path persistence, and a user-facing dashboard. Outline next steps: extended sensor coverage, predictive analytics on historical data, multi-tenant support, and hardening for production deployment.

Table I
HARDWARE COST PER SENSOR NODE

Component	Unit Cost
ESP32-DevKitC	8 €
Bosch BME680	18 €
TSL2561 Light Sensor	5 €
DHT11 Sensor	3 €
Joy-it KY-026 Flame Sensor	3 €
Joy-it KY-028 Thermistor	3 €
Prototyping equipment	~3 €
Total	~43 €

Future work notes: - Deep Sleep for ESP32 on Hardware for longer battery life - AI Outlier detection on AWS - Predictions on future sensor readings

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